HARMAN PREET SINGH

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ACADEMIC EMPLOYMENT

University of Tübingen *Nov. 2025 - Present*
Post-doctoral researcher in the groups of Marius Lemm and Stefan Teufel Tübingen - Germany

Scuola Internazionale Superiore di Studi Avanzati (SISSA) *Oct. 2021 - Oct. 2025*
Ph.D. student in the group of Marcello Porta Trieste - Italy

EDUCATION

Scuola Internazionale Superiore di Studi Avanzati (SISSA) *Oct. 2021 - Oct. 2025*
Ph.D. in Geometry and Mathematical Physics Trieste - Italy
`Advisor:` Marcello Porta (Mathematics Area)
`Thesis:` *Large-scale response theory for gapless lattice Fermi systems in low dimensions*
`Final Mark:` Ph.D. awarded cum laude

Università degli Studi di Padova *Oct. 2019 - Sep. 2021*
M.Sc. in Physics Padova - Italy
`Advisors:` Antonio Ponno & Lorenzo Zanelli (Mathematics Department)
`Thesis:` *A rigorous derivation of the Bogoliubov-Gross-Pitaevskii equation for superfluids*
`Final Mark:` 110/110 cum laude

Università degli Studi di Padova *Oct. 2016 - Sep. 2019*
B.Sc. in Physics Padova - Italy
`Advisor:` Pieralberto Marchetti (Physics Department)
`Thesis:` *Quantum Mechanics in phase space: non-classical properties*

PUBLICATIONS AND PREPRINTS

3. M. Lemm, H. Lu, and H. P. Singh, *Lieb-Robinson bound for one-dimensional continuum fermions*, arXiv: 2611.11655[math-ph], (2026)

2. M. Porta, G. Scola, and H. P. Singh, *Large scale dynamical response of interacting* 1*d Fermi systems*, arXiv: 2509.08665[math-ph], (2026) to appear on *Comm. Math. Phys.*

1. M. Porta, and H. P. Singh, *Large Scale Response of Gapless* 1*d and Quasi-*1*d Systems*, Annales Henri Poincaré (2025) https://doi.org/10.1007/s00023-025-01600-z

TEACHING EXPERIENCE

Mathematical Quantum Theory *WS 2026*
Teaching assistant for prof. M. Lemm, MSc. in Mathematical Physics program Tübingen

Advanced topics in Mathematical Quantum Theory *SS 2026*
Teaching assistant for prof. M. Lemm, MSc. in Mathematical Physics program Tübingen

Mathematical Quantum Theory *WS 2025*
Teaching assistant for prof. S. Teufel, MSc. in Mathematical Physics program Tübingen

CONFERENCE TALKS AND POSTER SESSIONS

Jan 2027 **Trails in Quantum Mechanics and Surroundings**, Napoli, Italy.
Invited speaker in session *Quantum transport and geometric methods for solid state physics.*

13/2/25 **Mathematical Challenges in Quantum Mechanics**, GSSI, L'Aquila, Italy.
Contributed talk: *Large scale edge response for 2d topological insulators*

16/12/24 **127th Statistical Mechanics Conference**, Rutgers University, Piscataway, New Jersey, USA.
Contributed talk: *Validity of edge linear response for 2d topological insulators*

29/8/24 **Rigorous Renormalization Group Analysis of Collective Phenomena in Fermionic Quantum Systems**, Como, Italy.
Contributed poster.

13/8/24 **Quantissima in the Serenissima V**, Venice, Italy.
Contributed talk: *Validity of edge linear response for 2d topological insulators*

SEMINAR TALKS

15/12/25 **Junior SFB Coffee Talk**, online seminar
Validity of Kubo formula for 1d and quasi-1d lattice fermionic systems

13/11/25 **Mathematical Physics Oberseminar**, University of Tübingen, Tübingen, Germany
Large-scale response theory for gapless lattice Fermi systems in low dimensions

30/1/25 **Mathematical Physics Oberseminar**, University of Tübingen, Tübingen, Germany
Large scale response of gapless 1d and quasi-1d fermionic systems

13/3/23 **Junior Geometry and Mathematical Physics Seminar**, SISSA, Trieste, Italy
A geometric picture of topological phases of matter

1/6/22 **Renormalization Group seminar**, SISSA, Trieste, Italy
Determination of the RG fixed point via tree expansion

31/5/22 **Seminar-exam**: Riemann surfaces and integrable systems, SISSA, Trieste, Italy
The canonical ensemble of the periodic Toda lattice

SERVICE ACTIVITIES

Member of **SFB Junior Committee** *Nov 2025 - Present*
within Collaborative Research Center TRR 352 *Mathematics of Many-Body Quantum Systems and Their Collective Phenomena*, joint venture of LMU and TU in München, and U. Tübingen

ORGANISATION OF EVENTS

Co-organiser of the monthly **SFB Junior Seminar** *Nov 2025 - Present*

Co-organiser of the Junior SFB Meeting, Teisendorf 9-13/11/26.

SCIENTIFIC VISITS

University of Zürich (UZH) *May 2024*
Hosted by: B. Schlein and M. Porta 1 week

University of Zürich (UZH) *March 2024*
Hosted by: B. Schlein and M. Porta 1 week

University of Zürich (UZH) *Nov.-Dec. 2023*
Hosted by: B. Schlein and M. Porta 1 week

GRANTS AND SCHOLARSHIPS

2021-2025: SISSA PhD Fellowship (4 full years)

LANGUAGES

Italian	Native
Punjabi	Native
English	Fluent (level C1 on the CEFR for Languages)
Hindi	Fluent
French	Basic knowledge

Last updated: Tübingen, 10 - 11 - 2026.